

DAY-3 ASSIGNMENT [DSA&UML]

UDAY CHIRRA

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10. Pancake Sorting

```
class Solution {
    public List<Integer> pancakeSort(int[] arr) {
        List<Integer> result = new ArrayList<>();

        for (int size = arr.length; size > 1; size--) {
            int maxIndex = 0;

            for (int i = 1; i < size; i++) {
                if (arr[i] > arr[maxIndex]) {
                    maxIndex = i;
                }
            }

            if (maxIndex == size - 1) {
                continue;
            }

            if (maxIndex != 0) {
                flip(arr, maxIndex);
                result.add(maxIndex + 1);
            }

            flip(arr, size - 1);
            result.add(size);
        }

        return result;
    }

    private void flip(int[] arr, int end) {
        int left = 0, right = end;

        while (left < right) {
            int temp = arr[left];
```

```
    arr[left] = arr[right];  
    arr[right] = temp;  
    left++;  
    right--;  
}  
}  
}
```

